Class Test 6 Marking Scheme

General principles

G1 **Justified steps only** Marks are given only for logically justified steps. A correct statement asserted without its reason (e.g. a derivative formula with no mention of the chain rule, or a limit with no argument) earns at most about half of the marks of that step.

G2 **Every valid route earns full credit** Any mathematically valid argument earns full marks. Alternative routes are mapped onto the same marking steps. Unusual but correct steps are credited and alternative arguments are read generously.

G3 **Slips** Ordinary slips (notation, copying, a dropped factor that does not affect the logic, a misnamed theorem) cost 0.5–1 mark each, are deducted in the step where they occur, and are never penalised twice. Purely cosmetic slips that cannot mislead cost nothing.

G4 **False load-bearing claims** A false claim on which the argument depends loses the marks of every step that rests on it (later steps that are independent of it are still credited).

G5 **Assuming the conclusion** Assuming the conclusion, or quoting a result equivalent to it, earns nothing for the steps it replaces

G6 **What may be quoted** May be quoted without proof: the definition of concavity and of the derivative, the chain rule, the Mean Value Theorem, Taylor's theorem with Lagrange remainder, limits preserve weak inequalities, continuity of second partials for C^2 functions, the definition of an open set, and any EARLIER part of this test (also in its obvious version for a convex domain A). In parts (c) and (d) the one-variable derivative characterisation of concavity may also be quoted: for a C^1 function g on an open interval, g is concave if and only if g' is non-increasing, and for a C^2 function, g concave implies $g'' \leq 0$. Results that appear later in this test, the derivative characterisations in part (b) (where they are essentially what has to be proved), and the multivariable Hessian test in part (d) may not be quoted.

G7 **"By symmetry / similarly"** Where a part has a component that follows by symmetry (in this test: interchanging x and y at the end of Route B2 or Route C2, or the case $y < x$ in an argument written for $y > x$), full marks are given for any genuine indication of how it is done (e.g. "interchanging x and y, which were arbitrary"), about 60% of that component for a bare "similarly", zero if it is omitted, and $1 - 2$ points are deducted for an actual mistake in the reduction.

Labels: lower-case labels (a1, a2, ..., b1, ...) are marking steps; upper-case labels (Route A1, Route C3, ...) are alternative valid approaches; every route is marked on the same steps.

Part (a) [24 marks]

Claim. Let $f : \mathbb{R}^n \to \mathbb{R}$ be concave, fix $x, y \in \mathbb{R}^n$, and define $\varphi : [0,1] \to \mathbb{R}$ by $\varphi(t) = f(tx + (1-t)y)$. Then φ is concave on $[0,1]$, for every choice of $x, y \in \mathbb{R}^n$.

Model solution (Route A1) and marking steps

a1 (Set-up and quantifiers). Fix $x, y \in \mathbb{R}^n$ and take arbitrary $s, t \in [0,1]$ and $\lambda \in [0,1]$; put **[3]**

$$z(t) = tx + (1-t)y$$

so $\varphi(t) = f(z(t))$. (No deduction just for not remarking that $\lambda s + (1-\lambda)t \in [0,1]$.)

a2 (Affine identity). **[10]**

$$\lambda z(s) + (1-\lambda)z(t) = (\lambda s + (1-\lambda)t)x + (\lambda(1-s) + (1-\lambda)(1-t))y = rx + (1-r)y = z(r)$$

with $r = \lambda s + (1-\lambda)t$. The coefficients must be shown (one displayed regrouping with correct coefficients suffices).

a3 (Concavity of f, translated to φ). **[8]**

$$\varphi(r) = f(\lambda z(s) + (1-\lambda)z(t)) \geq \lambda f(z(s)) + (1-\lambda)f(z(t)) = \lambda\varphi(s) + (1-\lambda)\varphi(t)$$

, by concavity of f at the two points $z(s), z(t)$ with weight $\lambda \in [0,1]$.

a4 (Conclusion). Since s, t, λ were arbitrary, φ is concave on $[0,1]$; this holds for every $x, y \in \mathbb{R}^n$. **[3]**

Total: $3 + 10 + 8 + 3 = 24$.

Alternative valid routes seen in scripts

- **Route A2.** Start from $\varphi(\lambda s + (1-\lambda)t)$, expand the argument

$$(\lambda s + (1-\lambda)t)x + (1 - \lambda s - (1-\lambda)t)y$$

and regroup it as $\lambda(sx + (1-s)y) + (1-\lambda)(tx + (1-t)y)$; then apply concavity of f. Regrouping = a2, concavity = a3.

- **Route A3.** Start from the two points $x_1 = t_1 x + (1-t_1)y$, $x_2 = t_2 x + (1-t_2)y$, apply concavity of f to $f(\theta x_1 + (1-\theta)x_2)$ and identify $\theta x_1 + (1-\theta)x_2 = t_0 x + (1-t_0)y$ with $t_0 = \theta t_1 + (1-\theta)t_2$. Identification = a2, concavity = a3.

Part (b) [24 marks]

Claim. Let $I \subseteq \mathbb{R}$ be an open interval and g a C^1 function on I. If g is concave on I, then $g(y) - g(x) \leq g'(x)(y - x)$ for all $x, y \in I$.

Model solution (Route B1) and marking steps

b1 (Set-up). Fix $x, y \in I$; the case $y = x$ is trivial; let $t \in (0, 1]$. The point $x + t(y - x) = ty + (1 - t)x$ lies in I (I is an interval). **[3]**

b2 (Concavity input). $g(ty + (1 - t)x) \geq tg(y) + (1 - t)g(x)$, i.e., **[8]**

$$g(x + t(y - x)) - g(x) \geq t(g(y) - g(x))$$

.

b3 (Divide by $t > 0$). **[5]**

$$\frac{g(x + t(y - x)) - g(x)}{t} \geq g(y) - g(x)$$

for all $t \in (0, 1]$ (dividing by $t > 0$ keeps the direction).

b4 (Limit $t \to 0^+$). With $h = t(y - x) \neq 0$, the left side equals **[8]**

$$(y - x)\frac{g(x + h) - g(x)}{h} \to (y - x)g'(x) \quad \text{as} \quad t \to 0^+$$

the right side in **b3** does not depend on t and weak inequalities are preserved, so

$$g'(x)(y - x) \geq g(y) - g(x)$$

Valid for both signs of $y - x$.

Total: $3 + 8 + 5 + 8 = 24$.

Alternative valid routes seen in scripts

- **Route B2.** Base point y: $g(y + \lambda(x - y)) - g(y) \geq \lambda(g(x) - g(y))$, divide by $\lambda > 0$, let $\lambda \to 0^+$ to get $g'(y)(x - y) \geq g(x) - g(y)$, then interchange the names x, y (legitimate because x, y are arbitrary; G7 applies to this interchange). Steps as in B1.
- **Route B3.** Mean Value Theorem + monotone derivative: $g(y) - g(x) = g'(c)(y - x)$ and $g'(c)(y - x) \leq g'(x)(y - x)$. Valid only if "$g'$ is non-increasing" is PROVED from concavity (b2 = that proof), b3 = MVT, b4 = the sign analysis for both $y > x$ and $y < x$. Quoting the monotonicity gives b2 = 0 (G5).

Part (c) [24 marks]

Claim. Let $A \subseteq \mathbb{R}^n$ be open and convex and F a C^1 function on A. If F is concave on A, then $F(y) - F(x) \leq \nabla F(x) \cdot (y - x)$ for all $x, y \in A$

Model solution (Route C1) and marking steps

c1 (The one-variable function). Fix $x, y \in A$. Define $J := \{t \in \mathbb{R} : x + t(y - x) \in A\}$. Then, J is an open interval containing $[0, 1]$: **[5]**

- $t = 0 \in J$ as $x \in A$
- $t = 1 \in J$ as $y \in A$
- $J \ni t : 0 < t < 1$ as A is convex.
- Hence, $[0, 1] \subset J$
- J is open: let $t' \in J$, then, $x + t'(y - x) \in A$. Since, A is open, there exists ε_A such that $B = \mathcal{B}_{\varepsilon_A}(x + t'(y - x)) \subset A$. Define $\varepsilon_J = \dfrac{\varepsilon_A}{2\|y - x\|}$, then, $x + (t' + \varepsilon_J)(y - x) \in B$, as $\|(x + t'(y - x)) - (x + (t' + \varepsilon_J)(y - x)\| = \|\varepsilon_J(y - x)\| = \varepsilon_J\|y - x\| < \varepsilon_A$. The same argument holds for $x + (t' - \varepsilon_J)(y - x)$, hence, $(t' - \varepsilon_J, t' + \varepsilon_J) \subset J$

Define $\psi(t) = F(x + t(y - x))$ on J; $\psi(0) = F(x)$, $\psi(1) = F(y)$. Using only $[0, 1]$ where (b) needs an open interval costs 2 marks; not checking $x + t(y - x) \in A$ costs 1 mark.

c2 (ψ is concave). Using an argument similar to part (a) (in its version for the convex set A), we get ψ is concave on J: **[5]**

- Take any $s, t \in J$ and $\lambda \in [0, 1]$ and define $r := \lambda s + (1 - \lambda)t$
- $r \in J$ as J is an interval by convexity of A
- Define $z(r) := ry + (1 - r)x$. Then $z(\cdot)$ is an affine map:

$$\begin{aligned} \lambda z(s) + (1 - \lambda)z(t) &= \lambda(sy + (1 - s)x) + (1 - \lambda)(ty + (1 - t)x) \\ &= (\lambda s + (1 - \lambda)t)y + (\lambda(1 - s) + (1 - \lambda)(1 - t))x \\ &= z(r) \end{aligned}$$

- Concavity of F at $z(s)$ and $z(t)$, both in A gives,

$$\psi(r) \geq \lambda\psi(s) + (1 - \lambda)\psi(t)$$

c3 (ψ is C^1 and $\psi'(0)$). As we have fixed x and y in A, the affine map, $z : J \to A$, defined by $z(t) = x + t(y - x)$ has a $1 \times n$ Jacobian matrix given by $\mathcal{J}_z(t) = (y_i - x_i)$. As, F is C^1 on A, $\dfrac{\partial F}{\partial x_i}$ exist and are continuous on A. Then, by the chain rule, **[7]**

$$\begin{aligned} \psi'(t) &= (\nabla F(z(t))^\top \mathcal{J}_z(t) \\ &= \nabla F(x + t(y - x)) \cdot (y - x) \\ &= \sum_{i=1}^{n} \frac{\partial F}{\partial x_i}(x + t(y - x))(y_i - x_i) \end{aligned}$$

As $\psi(t) := F(z(t))$, hence, $\psi'(t)$ is continuous in t, that is, ψ is C^1 on J so $\psi'(0) = \nabla F(x) \cdot (y - x)$. A componentwise computation counts as the chain rule; the bare formula earns about half (G1).

c4 (Apply (b) and conclude). Part (b) with $g = \psi$ at the points 0 (as x) and 1 (as y): $\psi(1) - \psi(0) \leq \psi'(0)(1-0)$, i.e. $F(y) - F(x) \leq \nabla F(x) \cdot (y - x)$, for all $x, y \in A$. **[7]**

Total: $5 + 5 + 7 + 7 = 24$.

Alternative valid routes seen in scripts

- **Route C2.** Parametrise from y: $g(t) = F(tx + (1-t)y)$; (b) at base point 0 gives $F(x) - F(y) \leq \nabla F(y) \cdot (x - y)$, then interchange x and y (G7 applies to this interchange); or (b) at base point 1 with the other point 0, which gives the claim directly. Steps as in the model.
- **Route C3.** Mean Value Theorem + quoted monotonicity of ψ' (allowed in (c) by G6): $\psi(1) - \psi(0) = \psi'(\xi)$ for some $\xi \in (0, 1)$, and $\psi'(\xi) \leq \psi'(0)$ because ψ' is non-increasing (ψ concave by (a)). Same steps: c1; c2 = concavity of ψ via (a); c3 = chain rule; c4 = MVT and the comparison $\psi'(\xi) \leq \psi'(0)$.

Part (d) [28 marks]

Claim. Let $A \subseteq \mathbb{R}^n$ be open and convex and F a C^2 function on A. If F is concave on A, then $v^T H_F(x) v \leq 0$ for all $x \in A$ and all $v \in \mathbb{R}^n$.

Model solution (Route D1) and marking steps

d1 (Fix x and an arbitrary v; room to move). Fix $x \in A$, $v \in \mathbb{R}^n$, $v \neq 0$. A is open: there is $r > 0$ with $\mathcal{B}_r(x) \subseteq A$, so $x + tv \in A$ for $|t| < \delta = r/\|v\|$ (or: $y = x + \varepsilon v \in A$ for small $\varepsilon > 0$, which reaches every direction v). **[5]**

d2 (First-order input). part (c) with x and $x+tv$ gives $F(x+tv) - F(x) \leq t \nabla F(x) \cdot v$. (Route D2: ψ' non-increasing, derived from (b) at two points (or quoted, G6);. Route D3: concavity gives $\psi(0) \geq \frac{1}{2}\psi(t) + \frac{1}{2}\psi(-t)$. Route D4: ψ is concave (by (a)), so by the quoted one-variable test $\psi'' \leq 0$ on the open interval). **[6]**

d3 (Second-order expression). Taylor (Lagrange form): **[7]**

$$F(x + tv) - F(x) = t \nabla F(x) \cdot v + \frac{1}{2} t^2 v^T H_F(z(t)) v$$

with $z(t) = x + (1 - \theta(t))tv$ for SOME $\theta(t) \in [0, 1]$. Routes D2/D3/D4: chain rule twice, $\psi''(t) = v^T H_F(x + tv) v$.

d4 (Combine). $\frac{1}{2} t^2 v^T H_F(z(t)) v \leq 0$, so $v^T H_F(z(t)) v \leq 0$ for $0 < t < \delta$. For other routes: **[5]**

D2 $\dfrac{\psi'(h) - \psi'(0)}{h} \leq 0$ and $h \to 0^+$ gives $\psi''(0) \leq 0$

D3 $\dfrac{\psi(t) - 2\psi(0) + \psi(-t)}{t^2} \leq 0$ and this quotient $\to \psi''(0)$ (needs a reason, e.g. Taylor or L'Hôpital)

D4 $\psi''(0) \leq 0$.

d5 (Pass to x and all v). As $t \to 0^+$, we have for D1: $\|z(t) - x\| \leq t\|v\| \to 0$ and $u \mapsto v^T H_F(u) v$ is continuous (F is C^2), so $v^T H_F(x) v = \lim_{t \to 0} v^T H_F(z(t)) v \leq 0$. D2/D3/D4: $\psi''(0) = v^T H_F(x) v \leq 0$ at the interior point $t = 0$. As x, v were arbitrary, $H_F(x)$ is negative semidefinite. **[5]**

Total: $5 + 6 + 7 + 5 + 5 = 28$.

Alternative valid routes seen in scripts

- **Route D2.** One variable via (b): $\psi(t) = F(x + tv)$; (b) at s and t, added, gives $(\psi'(t) - \psi'(s))(t - s) \leq 0$, so ψ' is non-increasing and $\psi''(0) \leq 0$; chain rule $\psi''(0) = v^T H_F(x) v$. (Also seen with $v = y - x$ followed by scaling $y = x + \varepsilon v$.) Quoting that ψ' is non-increasing instead of deriving it is allowed (G6).
- **Route D3.** Symmetric second difference: concavity of ψ on $(-\delta, \delta)$ gives $\psi(t) - 2\psi(0) + \psi(-t) \leq 0$; divide by t^2 and let $t \to 0$, the quotient tending to $\psi''(0)$ (with a reason); chain rule $\psi''(0) = v^T H_F(x) v$.

- **Route D4.** Quoted one-variable test (allowed in (d) by G6): $\psi(t) = F(x + tv)$ is concave by (a), so $\psi'' \leq 0$; chain rule $\psi''(0) = v^T H_F(x)v$. The concavity of ψ must be justified (by (a)). (Also seen with $v = y - x$ followed by scaling $y = x + \varepsilon v$.)